

Proposed Solution

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Team ID	SWTID-2026-8550
Project Title	AI BlogNest API

2. Proposed Solution

The proposed solution is ContentGen, a secure AI-powered REST API platform for blog creation and content management.

Component	Function
REST API	Structured application endpoints.
Express Server	Routes, HTTP methods, JSON and CORS.
Authentication	JWT-based stateless authentication.
Authorization	Admin, Editor, Author and Reader roles.
Services	Blog, comment and workflow business rules.
MongoDB + Mongoose	Persistent validated data.
AI Flow	Blog generation and summarization.
Security	Validation, sanitization, rate limiting and logging.

Functional Flow

API Client → Express Server Gateway → Authentication/RBAC → Service Modules → Mongoose → MongoDBAI requests → AI Flow → Google Gemini AI → Generated/Summarized Content.

API Capabilities

Registration and login.
Create/fetch/manage blogs.

Generate AI blogs.
Summarize blogs.
Categories and tags.
Comment moderation.
Media metadata handling.
Search and protected operations.